

THIRD SEMESTER DIPLOMA EXAMINATION IN
ENGINEERING AND TECHNOLOGY

BASIC MEDICAL SCIENCE – SET-1

MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	What is the flaw in the argument that “Organ system is the basic functional unit of living organism”	M1.02	U
2	What is resting membrane potential?	M1.04	R
3	What is the equipment used for blood pressure measurement?	M2.03	U
4	The range of normal breathing rate in human is	M2.04	U
5	Define vital capacity?	M2.05	U
6	Peripheral nervous system consist of and	M3.01	R
7	Functional unit of kidney is.....	M3.04	U
8	Name the two sensory receptors in eye?	M4.01	R
9	List the bones of middle ear?	M4.02	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the process of action potential generation in a cell?	M1.03	R
2	Draw the structure of cell and label its parts ?	M1.02	R
3	Describe the electrical activity of heart?	M2.03	U
4	Describe the mechanism of breathing?	M2.04	R
5	Explain the structure and function of mitochondria in a cell?	M1.02	R
6	List the 4 chambers of heart and its functioning?	M2.02	U
7	Explain the structure of urinary system?	M3.03	R
8	List the functions of cerebellum?	M3.01	U
9	Describe the process of sensation of smell?	M4.03	U
10	Explain the role of ear in maintaining the equilibrium of body?	M4.02	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain transmission of nerve impulse with suitable diagram?	M1.04	R
	OR		
IV	Explain basic anatomical positions and planes?	M1.01	R
V	Explain any 3 types of tissues and their functions?	M1.02	R
	OR		
VI	Describe how the membrane potential are recorded?		

		M1.03	U
VII	Draw the structure of human heart and explain ?	M2.02	R
	OR		
VIII	Illustrate the process of gas exchange in respiration process? .	M2.04	R
IX	Differentiate between systemic and pulmonary circulation?	M2.01	U
	OR		
X	Explain the anatomy of respiratory system?	M2.04	U
XI	Describe the physiology of urine formation ?	M3.03	R
	OR		
XII	Explain the general organization of Nervous system?	M3.01	R
XIII	Explain the sensory receptors in human skin	M4.03	R
	OR		
XIV	Describe the process of hearing	M4.02	R

THIRD SEMESTER DIPLOMA EXAMINATION IN
ENGINEERING AND TECHNOLOGY

BASIC MEDICAL SCIENCE-SET-2

MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	The resting membrane potential of a large nerve is.....	M1.03	R
2	Write one example for connective tissue.	M1.02	R
3	The concentration of Na ⁺ ion is more in fluid	M2.03	U
4	The range of vital capacity in normal human being is	M2.05	R
5	Define residual volume ?	M2.05	R
6	List the parts of Central Nervous system.	M3.01	R
7	Name the blood vessel entering the Bowmans Capsule?	M3.03	U
8	Name the parts of inner ear?	M4.02	R
9	List two types of cutaneous receptors?	M4.03	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the structure and function of muscle cell?	M1.02	R
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2	Draw the structure of nerve cell and label its parts ?	M1.02	R
3	Describe the cardiac cycle of heart?	M2.03	U
4	Describe the mechanism of breathing?	M2.04	R
5	Explain the structure and function of cell membrane in a cell?	M1.02	R
6	List the valves of heart and its functioning?	M2.02	U
7	Explain the structure of nephron?	M3.04	R
8	List the functions of cerebrum?	M3.02	U
9	Describe the process of sensation of taste?	M4.03	R
10	The colour of object is not clear at night. Explain the reason?	M4.01	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Classify the human tissues and mention the function of any three types?	M1.02	R
IV	OR Explain basic anatomical positions and planes?	M1.01	R
V	Explain any 3 types of cells and their functions?	M1.02	R
VI	OR Describe how the membrane potential are recorded?	M1.03	U
VII	Explain the major blood vessels of heart ? OR	M2.02	R

VIII	Illustrate the mechanism of respiration? .	M2.04	U
IX	Differentiate between systemic and pulmonary circulation?	M2.04	U
	OR		
X	Explain the function of heart?	M2.02	R
XI	Describe the anatomy of brain ?	M3.02	R
	OR		
XII	Explain the structure of Nephron?	M3.04	R
XIII	Explain the anatomy of eye with suitable diagram	M4.01	R
	OR		
XIV	Describe the process of sensing taste.	M4.03	R